Extended Case 5 — Mountain States Potato

Production diagnosis with multiple regression · 307312 Applied Statistics — Masters Level

- **Start after:** multiple regression and model diagnostics in Part 3.
- **Suggested time:** 90–120 minutes.
- **Business question:** Which measured production factors help explain filter-cake solids content, and what should management investigate?

Learning outcomes:

- Interpret adjusted regression associations in business units.
- Compare models while checking collinearity and influential measurements.
- Separate prediction, process intervention and evidence of causation.

1. The production manager's brief

- Filter cake is a potato-processing by-product used in cattle feed.
- Solids content had historically been slightly above 12%, but recent levels were nearer 11%.
- **Source:** *Basic Business Statistics*, ninth edition, Chapter 15, p. 623; variable dictionary also p. 790.
- **Unit:** one process measurement; 54 rows with solids and six potential predictors.
- **Period:** measurements were taken several times daily over 20 days; calendar dates and day IDs are absent from the workbook.
- **Limit:** observations may be temporally dependent; randomized process settings are not established.
- The management goal is to investigate how to return solids **above 12%**, not merely maximize fitted R-squared.

2. Import the process measurements

```
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
potato <- read.csv(file.path(data_dir, "business", "extended-potato.csv"))
stopifnot(nrow(potato) == 54, ncol(potato) == 7, !anyNA(potato))
knitr::kable(head(potato))
```

solids	ph	lower_pressure	upper_pressure	cake_thickness	varidriv_speed	drum_speed
9.7	3.7	13	14	0.250	6	33.00
9.4	3.8	17	18	0.875	6	30.43
10.5	3.8	14	15	0.500	6	34.00
10.9	3.9	14	14	0.500	6	34.00
11.6	4.3	17	18	0.375	6	36.24
10.9	4.2	16	17	0.500	6	31.76

- `data/business/extended-potato.csv`; guide: `data/business/README.md`.

- `solids`: percentage solids; `ph`: acidity.
- `lower_pressure` and `upper_pressure`: vacuum-line pressure below and above the fluid line.
- `cake_thickness`: thickness on the drum.
- `varidriv_speed`: control setting; `drum_speed`: measured rotation speed.
- The reviewed dictionary does not establish physical units for pressure, thickness or speed; retain source-scale labels.

3. Describe the gap from the target

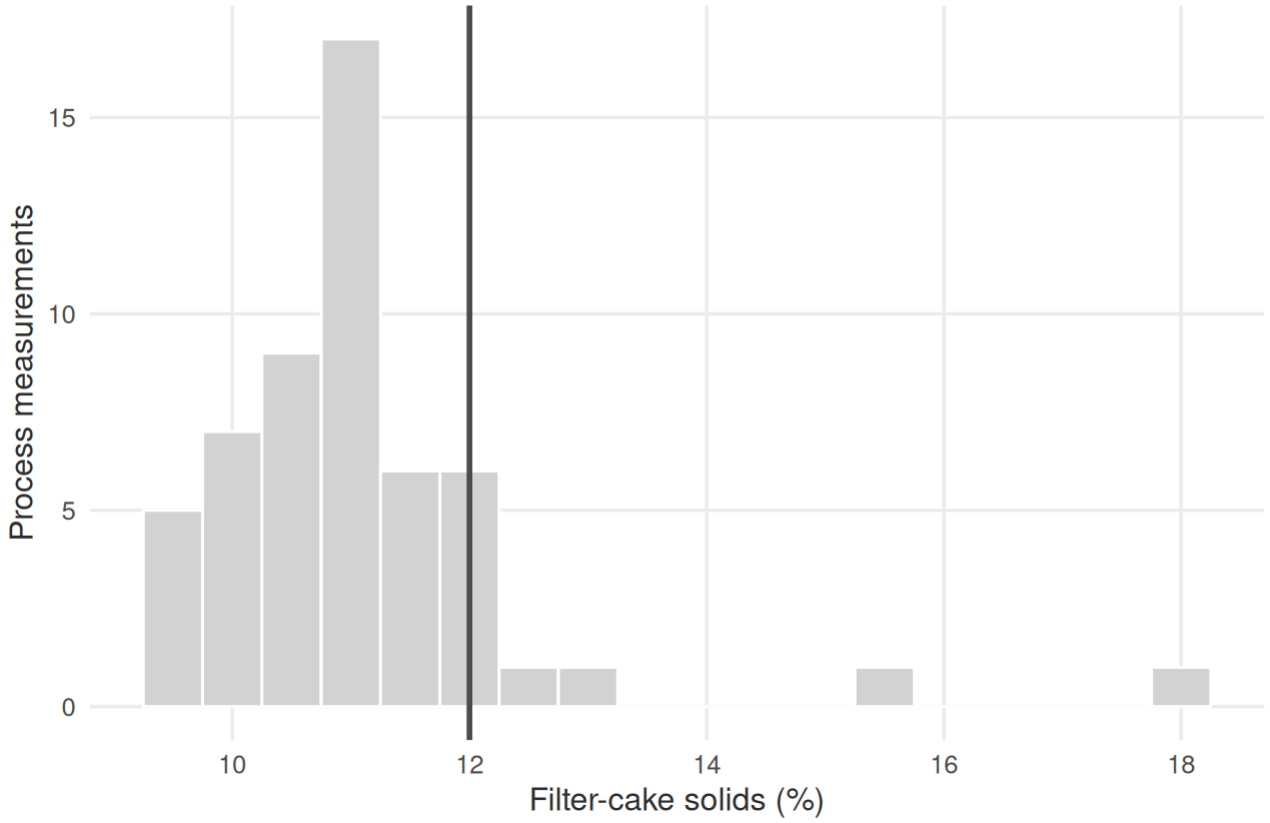
```
knitr::kable(data.frame(variable = names(potato),
  mean = sapply(potato, mean), sd = sapply(potato, sd),
  min = sapply(potato, min), max = sapply(potato, max)),
  row.names = FALSE, digits = 3)
```

variable	mean	sd	min	max
solids	11.093	1.373	9.40	17.800
ph	3.981	0.267	3.40	4.600
lower_pressure	14.426	1.987	8.00	18.000
upper_pressure	15.315	2.109	8.00	19.000
cake_thickness	0.438	0.131	0.25	0.875
varidriv_speed	5.778	0.538	4.00	6.000
drum_speed	38.211	6.328	30.43	60.400

```
knitr::kable(data.frame(measurements = nrow(potato),
  below_12_pct = 100 * mean(potato$solids < 12),
  at_or_below_12_pct = 100 * mean(potato$solids <= 12)), digits = 2)
```

measurements	below_12_pct	at_or_below_12_pct
54	90.74	92.59

```
ggplot(potato, aes(solids)) +
  geom_histogram(binwidth = .5, fill = petra_accent, colour = "white") +
  geom_vline(xintercept = 12, colour = petra_ink, linewidth = 1) +
  labs(x = "Filter-cake solids (%)", y = "Process measurements")
```



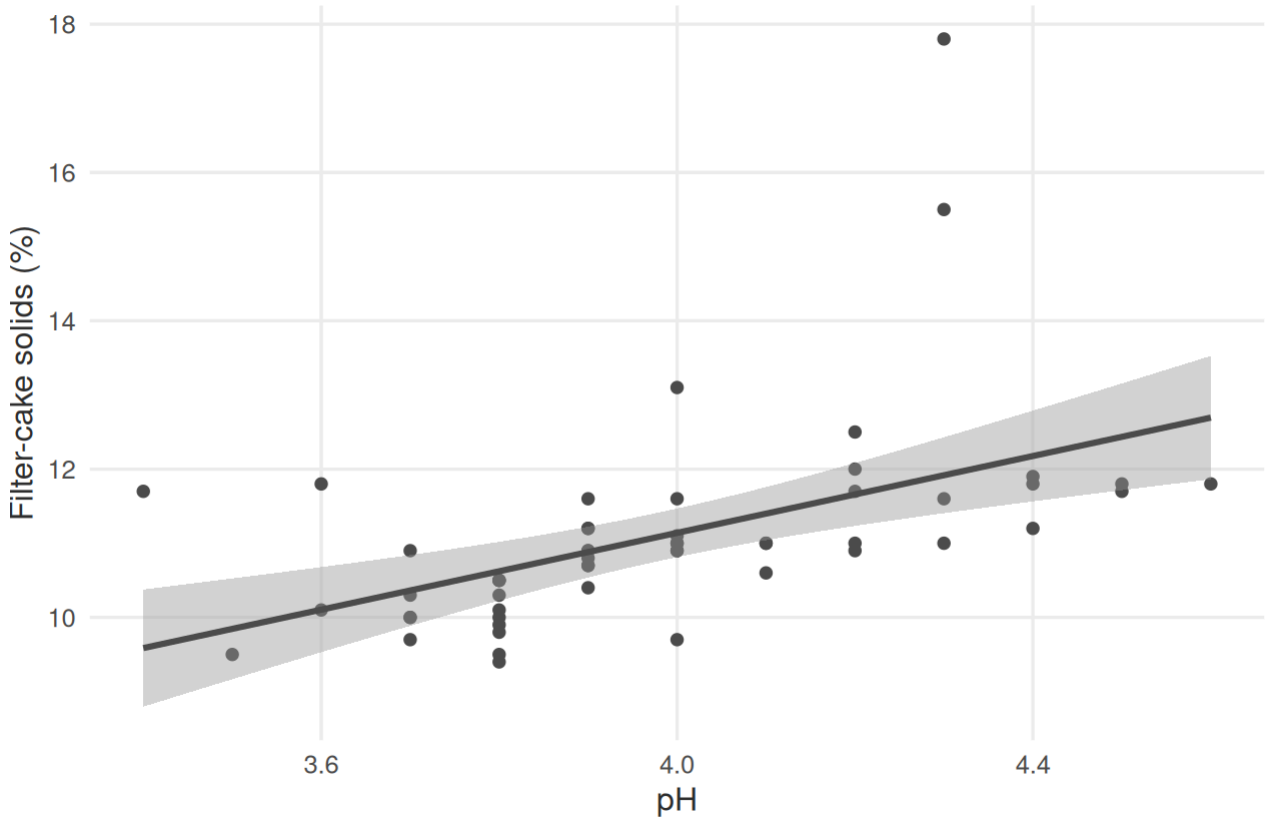
- The 12% line is a business reference, not an estimated statistical control limit.
- “Below 12%” and “not above 12%” use different comparisons for values exactly at 12.
- The percentages describe the 54 records; repeated measurements do not establish the plant-wide failure rate.

4. Explore associations and predictor overlap

```
knitr::kable(cor(potato), digits = 2)
```

	solids	ph	lower_pressure	upper_pressure	cake_thickness	varidriv_speed	drum_speed
solids	1.00	0.50	-0.34	-0.34	-0.28	0.25	-0.10
ph	0.50	1.00	0.12	0.12	-0.36	0.33	-0.29
lower_pressure	-0.34	0.12	1.00	0.94	0.44	-0.10	-0.03
upper_pressure	-0.34	0.12	0.94	1.00	0.44	-0.05	-0.09
cake_thickness	-0.28	-0.36	0.44	0.44	1.00	-0.18	0.12
varidriv_speed	0.25	0.33	-0.10	-0.05	-0.18	1.00	-0.94
drum_speed	-0.10	-0.29	-0.03	-0.09	0.12	-0.94	1.00

```
ggplot(potato, aes(ph, solids)) + geom_point(colour = petra_ink) +
  geom_smooth(method = "lm", formula = y ~ x, colour = petra_ink) +
  labs(x = "pH", y = "Filter-cake solids (%)")
```



- Correlation describes linear association; inspect plots before treating it as an adequate summary.
- Strong predictor correlations can make individual adjusted slopes unstable.
- Do not interpret a marginal pH association as a demonstrated process intervention effect.

5. Fit the full regression

```
full <- lm(solids ~ ph + lower_pressure + upper_pressure +
  cake_thickness + varidriv_speed + drum_speed, data = potato)
summary(full)
```

```
Call:
lm(formula = solids ~ ph + lower_pressure + upper_pressure +
    cake_thickness + varidriv_speed + drum_speed, data = potato)

Residuals:
    Min      1Q  Median      3Q     Max
-1.4287 -0.4711 -0.0692  0.3339  4.8631

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -13.50253    7.85428  -1.719   0.0922 .
ph               2.94422    0.64822   4.542 3.88e-05 ***
lower_pressure  -0.15421    0.21013  -0.734   0.4667
upper_pressure  -0.10619    0.19638  -0.541   0.5912
cake_thickness   1.50329    1.39232   1.080   0.2858
varidriv_speed   1.79546    0.83886   2.140   0.0375 *
drum_speed       0.14897    0.07001   2.128   0.0386 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.043 on 47 degrees of freedom
Multiple R-squared:  0.4885,    Adjusted R-squared:  0.4232
F-statistic: 7.482 on 6 and 47 DF,  p-value: 1.166e-05
```

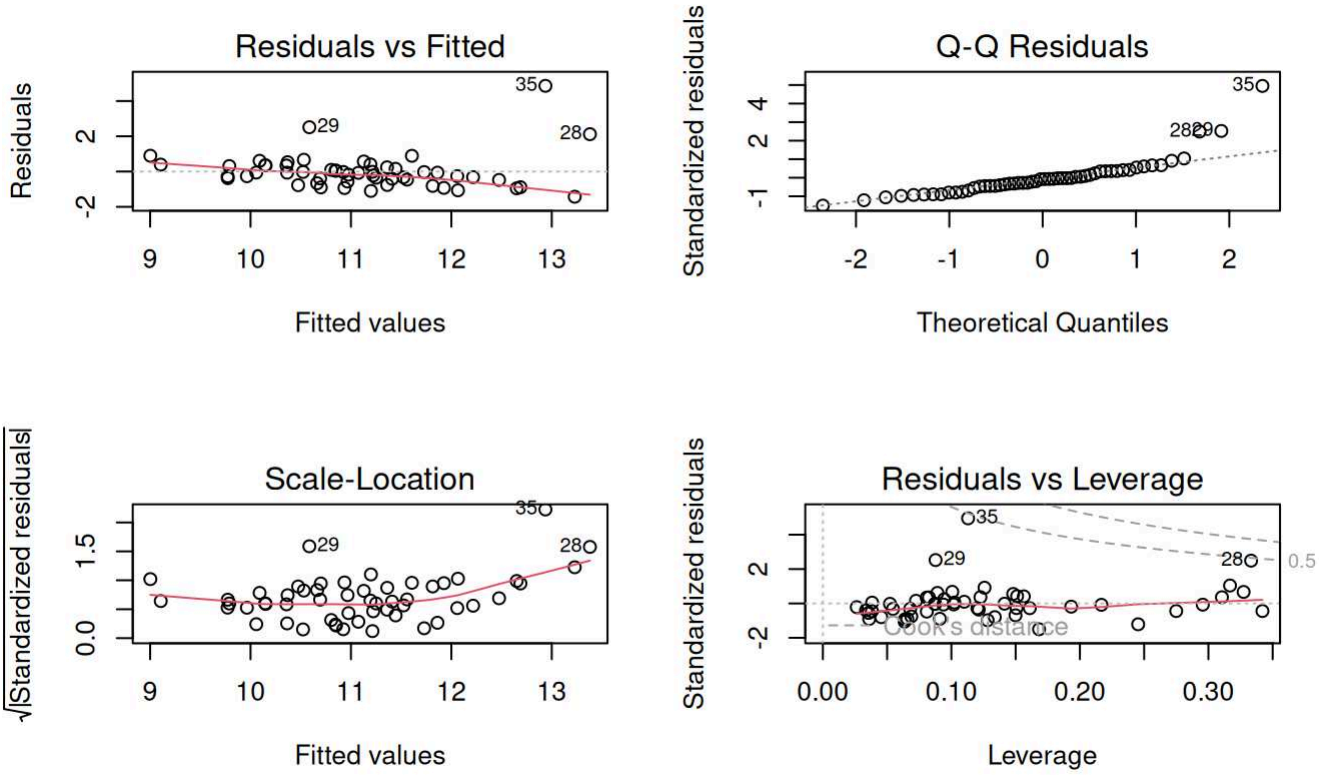
```r
confint(full)
```

```
                     2.5 %    97.5 %
(Intercept)    -29.303292556 2.2982419
ph               1.640157149 4.2482739
lower_pressure  -0.576947980 0.2685199
upper_pressure  -0.501247641 0.2888686
cake_thickness  -1.297697193 4.3042706
varidriv_speed   0.107902180 3.4830246
drum_speed       0.008117934 0.2898192
```

- Each slope describes the fitted solids difference holding the other included predictors fixed.
- With solids measured in percent, a one-unit predictor change corresponds to a **percentage-point** difference in solids.
- Conventional coefficient tests and intervals require suitable errors and sampling/model assumptions.
- The missing day/time identifiers limit checks of temporal dependence; nominal p-values are conditional, not validated process evidence.

6. Check diagnostics before using the model

```r
par(mfrow = c(2, 2))
plot(full)
```




```
par(mfrow = c(1, 1))
influence <- data.frame(source_row = seq_len(nrow(potato)),
  cooks_distance = cooks.distance(full), leverage = hatvalues(full))
knitr::kable(head(influence[order(-influence$cooks_distance), ], 5), digits = 3)
```

	source_row	cooks_distance	leverage
35	35	0.446	0.113
28	28	0.442	0.333
29	29	0.087	0.088
16	16	0.072	0.317
25	25	0.068	0.245

- Inspect curvature, changing residual spread, unusual residuals and influential rows.
- Row numbers identify records; they are not verified timestamps.
- A diagnostic flag is a reason to investigate, not an automatic rule for deleting a measurement.

A direct collinearity check

```
predictor_names <- names(potato)[names(potato) != "solids"]
vif <- sapply(predictor_names, function(variable) {
  others <- setdiff(predictor_names, variable)
  1 / (1 - summary(lm(reformulate(others, variable), data = potato))$r.squared)
})
knitr::kable(data.frame(predictor = names(vif), VIF = as.numeric(vif)), digits = 2)
```

predictor	VIF
ph	1.46
lower_pressure	8.49
upper_pressure	8.35
cake_thickness	1.61
varidriv_speed	9.92
drum_speed	9.56

- Each VIF compares a predictor with the other predictors; it measures variance inflation from their linear overlap.
- Large VIFs warn that holding the other predictors fixed may describe combinations with little support in the data.
- VIF does not assess omitted confounding or establish a causal model.

7. Compare a simpler model cautiously

```
reduced <- lm(solids ~ ph + cake_thickness + drum_speed, data = potato)
knitr::kable(AIC(full, reduced), digits = 2)
```

	df	AIC
full	8	166.30
reduced	5	179.67

```
summary(reduced)
```

```
Call:
lm(formula = solids ~ ph + cake_thickness + drum_speed, data = potato)

Residuals:
    Min      1Q  Median      3Q     Max
-1.3360 -0.6110 -0.1941  0.1124  5.7761

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.42148    3.42858   0.415 0.680211
ph             2.45269    0.69267   3.541 0.000873 ***
cake_thickness -1.22410    1.36387  -0.898 0.373743
drum_speed     0.01155    0.02746   0.421 0.675759
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.21 on 50 degrees of freedom
Multiple R-squared:  0.2679,    Adjusted R-squared:  0.2239
F-statistic: 6.098 on 3 and 50 DF,  p-value: 0.001282
```

- This reduced specification is a course illustration, not a textbook-selected model or a verified optimum.
- AIC balances fit and complexity for comparable models fitted to the same response and records.
- A lower in-sample AIC does not demonstrate reliable future prediction or causal effects.
- Process knowledge and validation on later, time-identified measurements are needed before operational use.

8. Independent production briefing

1. Report the mean solids and the percentages below, and at or below, 12%; explain the denominator.
2. Identify the strongest predictor overlaps using correlations and VIFs.
3. Interpret one coefficient from the full model in correct units, including what is held constant.
4. Compare full and reduced models using fit, diagnostics and process interpretability.
5. Investigate the most influential record. If you fit a sensitivity model without it, retain and report the original full-data model too.
6. Propose additional data collection with timestamps, day IDs and relevant operating conditions.
7. Design a controlled follow-up investigation of one promising factor within feasible operating ranges.

Submission

- One model-comparison table and two charts serving different questions.
- A 150–200-word executive summary distinguishing observed association, uncertain prediction and proposed controlled investigation.
- Do not recommend immediately changing plant settings based solely on these observational regressions.